Diagnosis of Ischemia-Causing Coronary Stenoses by Quantitative Coronary Angiography and LASSO Model

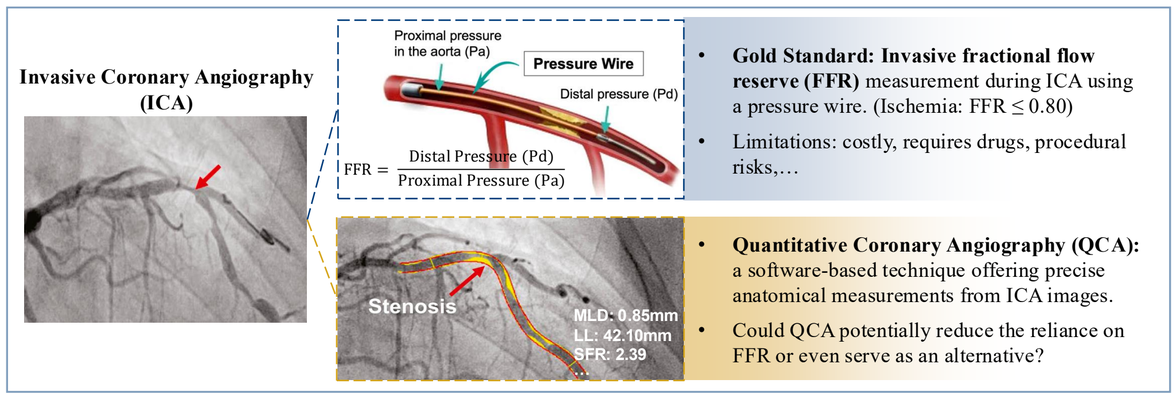
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Introduction

Background:

- Fractional flow reserve (FFR) ≤ 0.80 is considered the gold standard for evaluating functionally significant coronary stenosis. However, FFR has notable limitations: invasiveness, high cost, procedural associated risks.
- Quantitative coronary angiography (QCA) is a valuable imaging tool for assessing the severity of coronary artery disease (CAD) by detailed measurement of the lumen diameter. While QCA excels in the anatomical assessment of stenosis, its predictive ability for identifying hemodynamically significant stenoses is limited.
- Least Absolute Shrinkage and Selection Operator (LASSO) is a powerful method that performs regularization and feature selection, particularly useful in handling datasets with high dimensionality and multicollinearity.

Objective: The aim of this study was to determine the diagnostic performance of a **least absolute shrinkage and selection operator (LASSO)** model applied to quantitative coronary angiography (QCA) data and clinical data for diagnosing hemodynamically significant coronary stenosis in patients with suspected or known coronary artery disease (CAD).



Methods

Flowchart

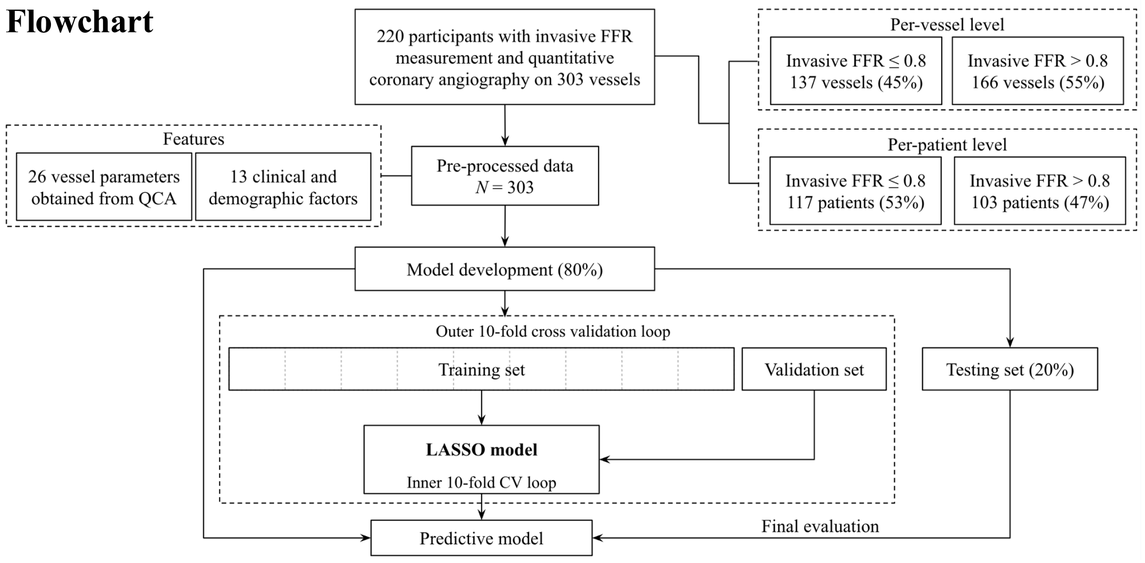


Figure 1. Flowchart of the study design and model development process. LASSO = least absolute shrinkage and selection operator, FFR = fractional flow reserve, QCA = quantitative coronary angiography, CV = cross validation.

LASSO Logistic Regression

To avoid overfitting and curse of dimensionality, we applied LASSO logistic regression to remove redundant features and find solutions with fewer non-zero coefficients. R package **glmnet** was used to implement the method.

- In logistic regression, the **negative log-likelihood** is given by:

$$-\sum_{i=1}^{n}\log(P_\beta(Y_i|X_i)) = \sum_{i=1}^{n}\{-Y_i(\sum_{j=0}^{p}\beta_j x^{(j)}) + \log(1+\exp(\sum_{j=0}^{p}\beta_j x^{(j)}))\}$$

- Written in terms of the **loss function ρ**, representing the negative log-likelihood for the i-th observation:

$$\rho_{(\beta)}(x,y) = -y(\sum_{j=0}^{p}\beta_j x^{(j)}) + \log(1+\exp(\sum_{j=0}^{p}\beta_j x^{(j)}))$$

- LASSO regularization works by adding an **ℓ1 penalty term** with a parameter λ to the log-likelihood function. The LASSO estimator is defined as:

$$\hat{\beta}(\lambda) = \underset{\beta}{\operatorname{argmin}}(n^{-1}\sum_{i=1}^{n}\rho_{(\beta)}(X_i,Y_i) + \lambda||\beta||_1)$$

where n is the number of observations, p is the number of features, $x_i^{(j)}$ is the value of the j-th feature for the i-th observation, β_j is the coefficient associated with j-th feature, $\|\beta\|_1$ is sum of absolute values of the coefficients.

Results

Feature Selection using LASSO Model

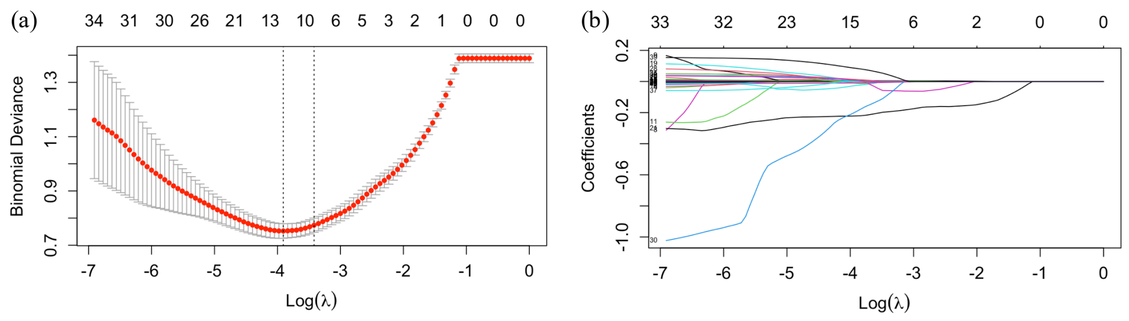


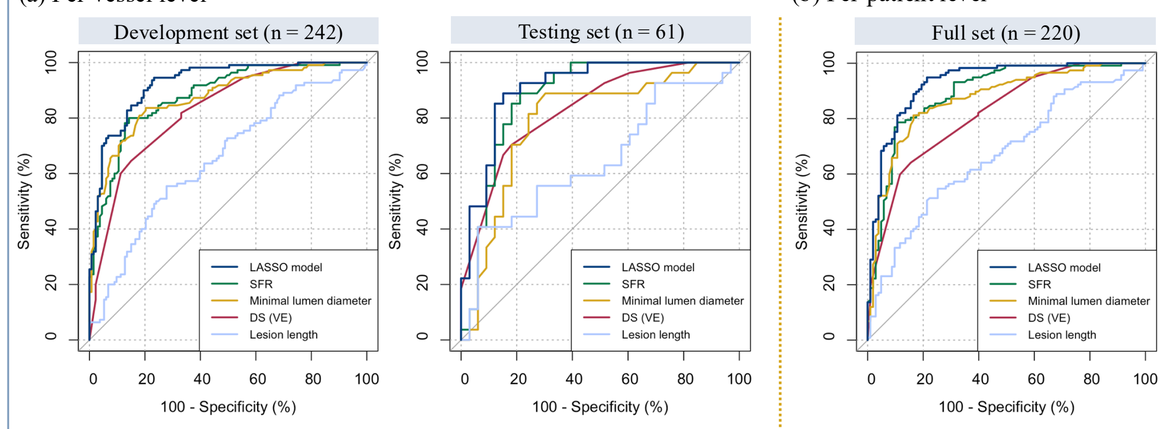
Figure 2. Feature selection by LASSO. **(a)** Binomial deviance versus log(λ). The optimal λ is determined by 10-fold cross validation and one standard error rule. The two dotted vertical lines represent the values at the minimum and one standard error above the minimum, respectively. **(b)** Coefficient profiles for all predictors plotted against log(λ).

Baseline Characteristics of Selected Predictors

Parameter	All (n=303)	FFR > 0.80 (n=166)	FFR ≤ 0.80 (n=137)	*P* Value (Group comparison)	Univariate Odds Ratio (95% CI)	*P* Value (OR)
DS (VE) (%)	70 (50, 80)	60 (50, 70)	80 (70, 80)	< 0.001	1.11 (0.08 to 1.13)	< 0.001
Minimal lumen diameter (mm)	1.27 (0.97, 1.6)	1.5 (1.28, 1.88)	0.97 (0.79, 1.21)	< 0.001	0.02 (0.01 to 0.04)	< 0.001
Lesion length (mm)	15.08 (10.36, 22.3)	13.05 (9.37, 18.4)	17.64 (12.02, 26.28)	< 0.001	1.06 (1.03 to 1.09)	< 0.001
SFR	3.23 (2.21, 4.08)	4 (3.32, 4.36)	2.24 (1.51, 2.84)	< 0.001	0.18 (0.13 to 0.26)	0.024
Outflow angle (degree)	11 (7, 17)	11 (7, 16)	12 (8, 19)	0.128	1.03 (1.00 to 1.06)	0.037
Characteristic	**All (n=220)**	**FFR > 0.80 (n=103)**	**FFR ≤ 0.80 (n=117)**	***P* Value (Group comparison)**	**Univariate Odds Ratio (95% CI)**	***P* Value (OR)**
Weight (kg)	67.9 (59.88, 76.2)	68.7 (60, 76.7)	66.5 (57, 75)	0.455	0.98 (0.97 to 1.00)	0.022
Height (m)	1.65 ± 0.08	1.65 ± 0.09	1.64 ± 0.08	0.588	0.06 (0.00 to 1.12)	0.061
Family history of CAD, n (%)	32 (15)	8 (8)	24 (21)	0.013	2.20 (1.18 to 4.24)	0.015

Note. FFR = fractional flow reserve, DS (VE) = diameter stenosis (visual estimation), SFR = stenotic flow reserve, CAD = coronary artery disease.

ROC Curve and Area Under the Curve Analysis



	AUC	95% CI
LASSO model	0.926	0.894 to 0.958
SFR	0.881	0.839 to 0.923
MLD	0.875	0.830 to 0.919
DS (VE)	0.828	0.779 to 0.878
Lesion length	0.654	0.585 to 0.723

	AUC	95% CI
LASSO model	0.910	0.836 to 0.984
SFR	0.878	0.786 to 0.970
MLD	0.791	0.671 to 0.913
DS (VE)	0.833	0.736 to 0.929
Lesion length	0.648	0.503 to 0.792

	AUC	95% CI
LASSO model	0.929	0.895 to 0.964
SFR	0.892	0.849 to 0.936
MLD	0.870	0.822 to 0.918
DS (VE)	0.819	0.765 to 0.872
Lesion length	0.669	0.598 to 0.740

Figure 3. Graph of receiver operating characteristic (ROC) curve analysis with area under the curve (AUC) on a **(a)** per-vessel basis (development set and testing set) **(b)** per-patient basis (full set). SFR = stenotic flow reserve, MLD = minimal lumen diameter, DS (VE) = diameter stenosis (visual estimation).

Performance Metrics and Confusion Matrices

(a)

	Cutoff	Accuracy	Sensitivity	Specificity	PPV	NPV
Per-vessel level						
LASSO model	0.407	84.8%	94.5%	76.7%	77.0%	94.4%
SFR	3.005	83.1%	80.0%	85.7%	82.2%	83.8%
Minimal lumen diameter (mm)	1.265	81.5%	83.6%	79.7%	77.3%	85.5%
DS (VE) (%)	70	75.7%	64.5%	85.0%	78.0%	74.3%
Lesion length (mm)	17.26	64.6%	55.4%	72.2%	62.2%	66.2%
Per-patient level						
LASSO model	0.404	87.3%	94.9%	78.6%	83.5%	93.1%
SFR	3.01	83.6%	78.6%	89.3%	89.3%	78.6%
Minimal lumen diameter (mm)	1.235	82.3%	81.1%	83.5%	84.8%	79.6%
DS (VE) (%)	70	73.6%	64.1%	84.5%	82.4%	67.4%
Lesion length (mm)	17.26	64.1%	54.7%	74.8%	71.1%	59.2%

(b1)

Predicted \ Actual	Positive	Negative
Positive	104 True Positive	31 False Positive
Negative	6 False Negative	102 True Negative

(b2)

Predicted \ Actual	Positive	Negative
Positive	111 True Positive	22 False Positive
Negative	6 False Negative	81 True Negative

Note. (a) Performance metrics for LASSO model and strong predictors at per-vessel and per-patient levels. **(b)** Confusion matrices for LASSO model at **(b1)** per-vessel and **(b2)** per-patient levels. PPV = positive predictive values, NPV = negative predictive values, SFR = stenotic flow reserve, DS (VE) = diameter stenosis (visual estimation).

Nomogram

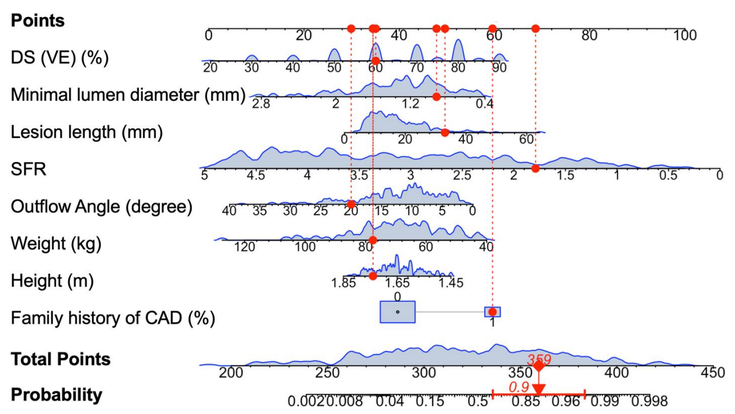


Figure 4. Individual nomogram for the prediction of hemodynamically significant stenosis with eight selected features. The nomogram results for one patient (red dots) are illustrated with an estimated probability of 90%. DS (VE) = diameter stenosis (visual estimation), SFR = stenotic flow reserve.

Note. The nomogram was built based on a **standard logistic regression** rather than the **LASSO logistic regression**. As LASSO model applies a penalty to the coefficients, it does alter the probabilities, potentially leading to underestimation or overestimation of true probabilities. Here we assume standard logistic regression provides "true" probabilities.

Conclusions

- **Summary:** The following parameters were selected for LASSO model: diameter stenosis, minimal lumen diameter, lesion length, stenotic flow reserve (SFR), outflow angle, weight, height, family history of CAD. On a per-vessel basis, the accuracy, sensitivity, specificity, PPV and NPV were 85%; 95%, 77% 77% and 94% (an AUC value of 0.926). On a per-patient level, accuracy, sensitivity, specificity, PPV and NPV were 87%, 95%, 79%, 84% and 93% (an AUC value of 0.929).
- **Conclusion:** The model showed a good diagnostic performance for predicting hemodynamically significant coronary stenosis on invasive coronary angiography.

References

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